DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSGD323
COURSE TITLE	**:**	Computer Animation and Simulation
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-CSGD311
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Mr. Jan Daenell Bancud
CONSULTATION HOURS	**:**	W/1000-1100/COS 100-C
CONTACT INFORMATON	**:**	jnbancud@dlsud.edu.ph 09524869736

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course introduces the essential workflows and techniques for creating 3D computer graphics, focusing on applications in architecture, product design, robotics simulations, and computer games. At the end of the course, students will identify key 3D components and workflows and gain proficiency in creating and animating 3D graphics using Autodesk Maya.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. identify key components of 3D modeling, animation, and simulation in Autodesk Maya and other industry-standard tools.

CLO2. utilize industry-standard workflows for modeling, texturing, lighting, and rendering while applying best practices for asset optimization, file management, and interoperability across 3D software platforms.

CLO3. examine ethical considerations, teamwork, and professional standards in 3D production.

CLO4. develop 3D assets and animations applicable to architecture, product development, robotics, and game design.

CLO5. assess the impact of resource-efficient modeling techniques and promote responsible use of computational and material resources; and

CLO6. construct a fully animated scene utilizing self-created 3D models, demonstrating proficiency in animation principles, rigging, rendering, and editing.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Identify key components of 3D modeling, animation, and simulation in Autodesk Maya and other industry-standard tools.	**TLO1.** Define computer animation and simulation while recognizing computer graphics elements **TLO2.** Discuss and elaborate how to navigate effectively in a 3D environment and use the software to create basic 3D shapes and modify them.
CLO2. Utilize industry-standard workflows for modeling, texturing, lighting, and rendering while applying best practices for asset optimization, file management, and interoperability across 3D software platforms.	**TLO3.** Apply mathematical foundations, functions and algorithms, such as extrusion, division, subdivision and sculpting, to increase or decrease polygon count.
CLO3. Examine ethical considerations, teamwork, and professional standards in 3D production.	**TLO4.** Demonstrate ethical practices and professionalism in teamwork by collaborating effectively with peers and colleagues on 3D modeling projects **TLO5.** Explore and explain the sustainable and responsible use of resources in 3D modeling, incorporating ethical practices

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus and Course Requirements **Module 0: Course Introduction**	Schoolbook	✓		2
Module 0	Week 2 Jan 27-Feb 1	- Lunar New Year (Holiday)	Schoolbook			0
Module 1 CLO1 TLO1	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer • Introduction to Computer Graphics and Simulation • Defining Computer animation and simulation • Key differences between Computer Graphics and Simulation **Module 1 - Introduction to Computer Graphics**	Schoolbook	✓		2

Module 1 CLO1 TLO1	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Other Computer Graphics Tools	Schoolbook	✓		2
Module 2 CLO2 TLO3	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • History of Computer Graphics • Elements of Computer Graphics	Schoolbook	✓		2
Module 2 CLO2 TLO3 CLO3 TLO4	Week 6 Feb 24-Mar 1	• Gospel Reading, Reflection, and Prayer • Teamwork and Collaboration with Maya **Module 2 - Advanced Modelling Techniques**	Schoolbook	✓		2
Module 2 CLO2 TLO3 CLO3 TLO4	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer • Polygonal Mathematical operations • Enabling Assessment 1. Short Quiz - History of Computer Graphics • Class Participation 1 Create a Workflow that works for you	Schoolbook	✓		2

Module 2 CLO2 TLO3 CLO3 TLO4	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer • How to increase and decrease polygon count • Enabling Assessment 2. Short Quiz - Maya Interface • Class Participation 2 Research on ways to optimize your model	Schoolbook	✓		2
Midterm Exam	Week 9 Mar 17-22	• Gospel Reading, Reflection, and Prayer • Midterm Summative Assessment on Modules 1-4		✓		2
TOTAL				10	8	18

Learning Outcomes for Final Period

CLO4. Develop 3D assets and animations applicable to architecture, product development, robotics, and game design.	**TLO6.** Apply various modeling techniques to design a simple house,while operating the tools and methods introduced. **TLO7.** Analyze and train robotic systems using Mimic for Maya.
CLO5. Assess the impact of resource-efficient modeling techniques and promote responsible use of computational and material resources.	**TLO8.** Identify how to use materials, textures, and basic shaders to industrial and product design models.
CLO6. Construct a fully animated scene utilizing self-created 3D models, demonstrating proficiency in animation principles, rigging, rendering, and editing.	**TLO9.** Create and design a 3D character, including the body, face, limbs, and hands, while incorporating industy standard practices **TLO10.** Determine how to animate characters or objects using the dope sheet, timeline, and IK techniques.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 3 CLO3 TLO5 CLO4 TLO6 TLO7 TLO8	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • **Sustainability Week** • Completion of Midterm Assessment • How to optimize polygon count • How to apply materials in Maya **Module 3 - Practicality of Modelling**	Schoolbook		✓	2
Module 3 CLO3 TLO5 CLO4 TLO6 TLO7 TLO8	Week 11 Mar 31-Apr 5	• Gospel Reading, Reflection, and Prayer • 3D Printing Materials and Sustainable product design • Enabling Assessment 1. Short Quiz - on Module 3	Schoolbook		✓	2
Module 3 CLO3 TLO5 CLO4 TLO6 TLO7 TLO8	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Using Maya for basic Architecture • How to install Mimic for Maya • Class Participation 1 Installing Mimic for Maya	Schoolbook	✓		2
	Week 13 Apr 14-19	***Independent Learning Days (April 14-15)*** ***Holy Week (April 16-19)***			✓	2

	Week 14 Apr 21-26	***Research Week***		✓	2
Module 4 CLO6 TLO9 TLO10	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • Creating a low poly character in Autodesk Maya **Module 4 - Animation and Rigging**	Schoolbook	✓	2
Module 4 CLO6 TLO9 TLO10	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer • Animation tools and workflow • Enabling Assessment 2. Short Quiz – On Module 4	Schoolbook	✓	2
Module 4 CLO6 TLO9 TLO10	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer • How to rig a character • Class Participation 2 Character Rigging	Schoolbook	✓	2
Final Exam	Week 18 May 19-24	• Gospel Reading, Reflection, and Prayer • Final Term Summative Assessment – Presentations of Final Project		✓	2
TOTAL			10	8	18

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Call Number/ E-provider	Reference Material
Journal	Kim T, Darke AM. 2025. Curly-cue: geometric methods for highly coiled hair. ACM Transactions on Graphics 44(2):1–12. Available online: https://dl.acm.org/doi/10.1145/3680528.3687641
Journal	Zhao F. 2024. A novel computational paradigm for reconstructing solid CAD features from a segmented manifold triangular mesh. Applied Sciences 14(14):6183. Available online: https://www.mdpi.com/2076-3417/14/14/6183
Journal	IEEE Transactions on Visualization and Computer Graphics. 2020. IEEE Trans. Vis. Comput. Graph. 26(1):1–2. Available online: https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=2945
Online	Autodesk. 2024. Maya Learning Channel. Available Online: **(159) Maya Learning Channel - YouTube**
Online	Autodesk. 2025. Maya 2025 Help: What's New in Maya 2025.2. Available from: https://help.autodesk.com/view/MAYAUL/2025/ENU/
Online	Autodesk. 2025. Maya Product Enhancements, Benefits & Roadmap. Available from: https://www.autodesk.com/customer-value/me/maya
Online	Autodesk. 2024. *Maya User's Guide*. Available from: https://help.autodesk.com/view/MAYAUL/2025/ENU/
SDC Publications	Murdock KL. 2023. Autodesk Maya 2024 Basics Guide. SDC Publications. Available From https://www.sdcpublications.com/Textbooks/Autodesk-Maya-2024-Basics-Guide/ISBN/978-1-63057-580-9/

Prepared: **Computer Science Department**
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Approved: **Josephine T. Eduardo**
Chair, CSD

Endorsed: **Marivic R. Mitschek**
Dean, CICS